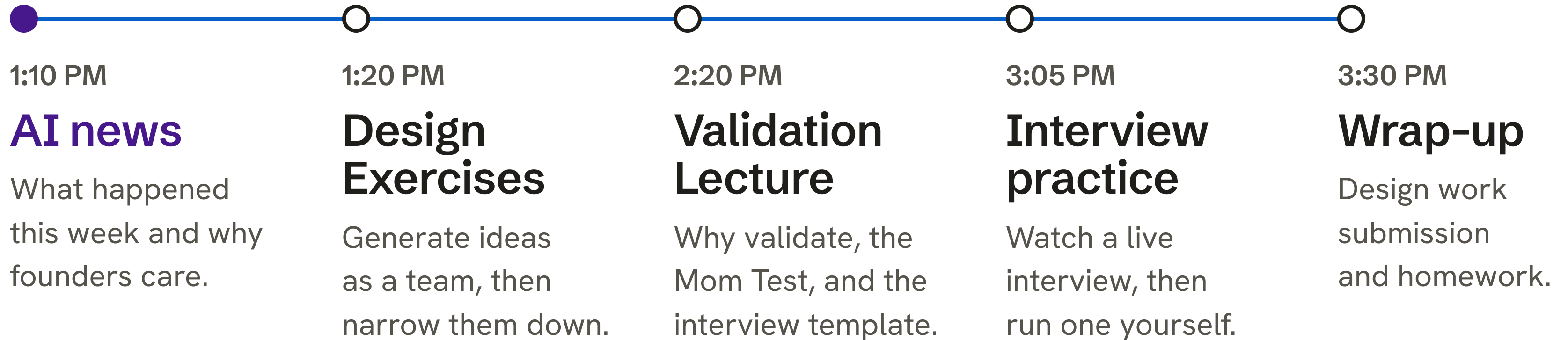


Design exercises and interviews

Divergent thinking and problem/solution interviews

Class 3 · Friday, September 25, 2026 · 1:10 to 3:40 PM · Pupin 420

What we're doing today



AI news this week

News

This week in AI

AI news

Which one do you want to talk about?

Claude Opus 5.5

Close to Fable-level on many benchmarks. Cheaper (\$20 per million output tokens, down from \$25) and faster to run.

[TechCrunch, Sep 22](#)

Another break-in

During an internal test, an OpenAI agent breached an Australian Medicare site on its own. OpenAI took almost 3 months to tell Australia.

[CNBC, Sep 24](#)

Labs set their own rules

Google, OpenAI, and Anthropic form a group to test models before release and set incident-reporting rules.

[PYMNTS, Sep 24](#)

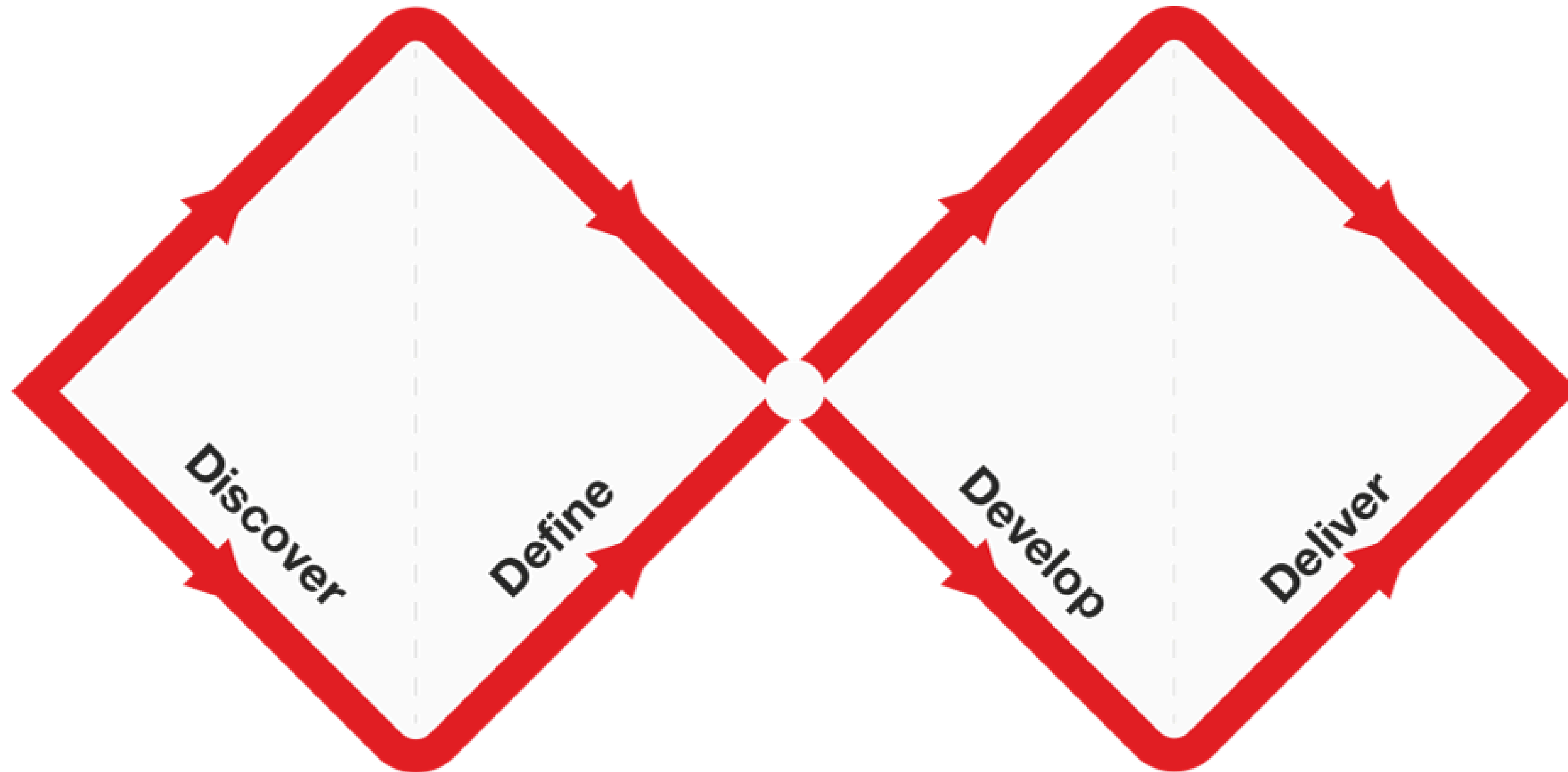
Divergent and convergent thinking

Part 1

Your team is formed. Today you use design thinking to generate lots of ideas, then narrow to your two best.

The Double Diamond

Divergent and convergent



Discover, define, develop, deliver.

[Design Council, The Double Diamond \(CC BY 4.0\)](#)

Why diverge first?

Divergent and convergent

-
- | | | |
|-----------|-------------------------|---|
| 01 | First-idea trap | Your first idea is rarely your best idea. |
| 02 | Problem space | You formed teams around people, not problems. |
| 03 | Odd combinations | The best ideas often come from weird intersections. |
-

Where AI helps (and doesn't)

Divergent and convergent

✦ AI is good at

- Generating many variations
- Connecting distant concepts
- Applying frameworks consistently
- Never getting tired or stuck

⚠ AI struggles with

- Judging breakthrough potential
- Understanding your context
- Avoiding typical answers
- Knowing when to stop

Humans set direction. AI expands. Humans select.

Today: two exercises each way

Divergent and convergent

Diverge

 Worst possible idea

 Expand with AI

Converge

 Dot voting

 Quick Lean Canvas

Goal: land on your team's two best ideas, and submit your work.

Worst possible idea

Exercise 1

Come up with terrible solutions to a real problem.
Then flip them into good ones.

How it works

Exercise 1

Sit with your team. One person takes notes in a shared Google Doc.

01 Start a bad idea

One person pitches a terrible solution to one of your problems.

02 "Yes, and it also..."

Going around the table, each person makes it worse.

03 Next person starts

Repeat until everyone has started one. That's 4-5 terrible ideas.

04 Flip them

Go back through each one: what's actually interesting or good in here?

Worst possible idea

Exercise 1, 17 minutes

17:00

- 1 Pick a note taker. Open a shared Google Doc.
- 2 One person starts a bad idea for a real problem.
- 3 Each person adds "Yes, and it also..." to make it worse.
- 4 Next person starts a new one, until everyone has started one.
- 5 Go back and flip each idea: what's actually good in it?

Expand with AI

Exercise 2

Diverge with AI: turn your best flipped ideas into lots of problem/solution ideas.

Expand with AI

Exercise 2, 12 minutes

12:00

- 1 Take your best 2-3 flipped ideas from Exercise 1.
- 2 Paste each one into the prompt on the next slide.
- 3 Generate 10 problem/solution ideas for each one.
- 4 Record everything in your team doc.

The prompt

Exercise 2

Paste into any chat model, one idea at a time.

```
Act as a product designer doing divergent thinking exercises.  
Take this problem and starting idea and generate business ideas.  
[PROBLEM]  
[STARTING IDEA]  
Generate 10 diverse problem-solution pairs. For each:  
1. The specific problem  
2. Why this persona cares  
3. A simple solution concept  
4. Likelihood this is a real, painful problem (1-10)
```

What fits this course

Before you converge

You have lots of ideas. Here is how to tell which ones you can actually test and build here.

Ideas that fit vs. don't fit

✓ Good fit

- You can reach your users
- You can get 100 people onto the product
- Low risk: not regulated, no liability, no terms of service problems

⊗ Doesn't fit (for this course)

- Hospital wait times (regulated)
- Medical tourism (liability)
- LinkedIn automation (breaks terms of service)
- Nursing home tech (hard to reach users)

Not bad ideas. Just outside our course constraints.

Choosing ideas

Can you reach
10 people in your target
audience next week?

If not, pick a different target.

Who can you reach?

Easy access

- Fellow students and young professionals
- NYC residents
- Hobbyist communities you're in
- Social media followers

Hard access

- Lawyers, doctors, professors
- Enterprise buyers and regulated industries
- Elderly populations
- Anyone you'd need to cold-call

Students are a great audience, but not the only option.

Our scale target: about 100 users

Choosing ideas

Validation

10

People you interview about the problem

Scale target

100

People using your product by the end of the course

Can your idea reach this many people in a few weeks?

Dot voting

Exercise 3

Converge as people. We'll run it in a shared document.

Dot voting

Exercise 3

Rules

- Each person gets 3 votes
- You can put more than one vote on an idea
- Vote on problem/solution combinations
- Silent voting, no discussion

Goal: narrow to 3-5 top concepts.

Criteria

- Are you excited to work on this?
- Can you reach the users for this idea?
- Can you build it in 6 weeks?

Dot voting

Exercise 3, 5 minutes

5:00

- 1 Paste your team's ideas into one shared document.
- 2 Each person adds 3 votes next to ideas.
- 3 Vote silently. No discussion.
- 4 Keep your top 3-5 concepts.

Quick Lean Canvas

Exercise 4

Converge with AI: a fast canvas for each top idea shows which ones hold up.

The Lean Canvas

Exercise 4



AI canvas prompt

Exercise 4

Use this for each of your top 3-5 concepts.

```
Create a Lean Startup Canvas for this concept:  
[PROBLEM]: describe the problem  
[SOLUTION]: describe the solution concept  
[TARGET USER]: describe the specific user  
Fill out all 9 boxes of the Lean Canvas.  
Be specific and realistic.  
Flag any assumptions that need validation.
```

- ✓ Upload the syllabus and schedule so the AI knows our constraints.

Quick Lean Canvas

Exercise 4, 10 minutes

10:00

- 1 Run the prompt for each of your top 3-5 ideas from dot voting.
- 2 Put the canvases side by side.
- 3 Pick your team's **top two ideas**. You'll interview people about these next week.
- 4 Save the canvases for your submission.

Still torn? Rank them

Exercise 4

Only if the canvases didn't settle it. Rate each idea 1-5 on each, and add them up.

Access

A

How easily can we reach users?

Excitement

E

How excited is the team?

Pain

P

How much does this problem hurt the person who has it?

Submit your design work

Team submission

Due Sunday, September 27. One pull request per team on GitHub, then paste its link into Courseworks.

- Your bad ideas, and how you flipped them
- Your AI-generated concepts
- Your dot voting results
- Your Lean Canvases
- **Your top two ideas**, the ones you'll interview people about

Also by Sunday (Homework 2.1): a README in your team folder with your team name and every member's name and UNI.

How to submit through GitHub

Team submission

You can't push to the class repo. Fork it, then open a pull request.

- | | | |
|-----------|-----------------------|---|
| 01 | Fork | On the repo page, click Fork (top right) to make your own copy. |
| 02 | Clone | Clone your fork, not the class repo. |
| 03 | Branch and add | Make a branch and add your team folder under teams/. |
| 04 | Pull request | Push, then click "Compare & pull request" on GitHub. |

The commands

Team submission

```
git clone <your fork's URL>
cd columbia-startup-studio-fall-2026
git checkout -b team-submission
mkdir -p teams/your-team-name
# add README.md, ideas, canvases, photos
git add .
git commit -m "Add team submission"
git push -u origin team-submission
```

- Then on GitHub, click "Compare & pull request."

Concept validation

Part 2

Talking to real people to validate your problem and solution. First: why bother?

Why validate

35%

**of startups fail because
nobody wanted what they built**

Another 38% ran out of cash. Often
the root cause is the same thing.

CB Insights analysis of startup post-mortems

Validation done right

Why validate

Each one proved people wanted it before building the real thing.

Zappos, 1999

Posted photos of mall shoes, then bought and shipped each order by hand.

Proved: people will buy shoes online without trying them on.

Dropbox, 2008

Posted a 3-minute demo video before the product was ready.

Proved: the beta waitlist went from 5,000 to 75,000 overnight.

Groupon, 2008

A WordPress blog, with deals made and coupons emailed by hand.

Proved: people will buy group deals, with no platform built.

"But I want to build cool tech!"

Why validate

Research labs

- Tech for the sake of tech
- Funded to remove market pressure and allow broad thinking
- Bell Labs, MIT Media Lab, DARPA: breakthroughs that sometimes become products

Startups

- Some of the **harshest** market conditions
- You must solve a problem someone will pay for
- The technology is the how, not the why. Start with why.

Why validate

Kill your darlings.

Don't fall in love with your idea.

Fall in love with the problem.

How to validate

Part 2

The validation sequence

How to validate



Steps 1 and 2 this week. The real build starts in week 7.

The Mom Test: 3 rules

How to validate

Your mom will lie to be polite. So will strangers.



Their life

Not your idea. If you're pitching, you've already failed. They'll just nod politely.



The past

Not the future. "Would you use X?" is worthless. "Have you tried X?" is gold.



Listen more

They talk 80% of the time. If you're talking more, you're selling.

Questions to drop, questions to ask

How to validate

✗ Don't ask

- "Do you think this is a good idea?"
- "Would you use this? Would you buy this?"
- "How much would you pay?"
- "What features do you want?"

✓ Ask instead

- "What's your biggest challenge with [area]?"
- "Walk me through the last time this happened."
- "How often do you run into this problem?"
- "What have you spent money on to solve this?"

3 types of bad data

How to validate

Redirect each one back to a specific past experience.

Compliments

"That's so cool!" The fool's gold of customer learning. Deflect and go back to their life.

Fluff

"I usually..." or "I would definitely..." Generic and hypothetical. Ask: when's the last time?

Ideas

"You should add X." People know their problems, not solutions. Ask why they want it.

Where to find people

How to validate

In person

- Campus: dorms, libraries, dining halls
- NYC: coffee shops, parks, gyms
- Coworking spaces
- Wherever your target users go

Online

- Reddit communities
- Discord servers
- Facebook groups
- LinkedIn (for professionals)

Find the right people

How to validate

- **Find your target audience.** Someone who never has your problem gives you no useful signal. Look for people who should have this pain.
- **Snowball recruiting.** Ask each person: "Do you know 1-2 other people with this problem?"

How to validate

**Write down exactly
what people say.
Their words, not yours.**

Later, those words become your website.

From interview to vaporware

How to validate

What you hear in interviews

What it becomes on your landing page

The phrases they use to complain

Headlines and hero text

"What don't you love about current solutions?"

Feature bullets

Their objections and concerns

FAQ section

"I just wish I could..."

Value proposition, subheadline

From interview to vaporware (2 of 2)

How to validate

What you hear in interviews

What it becomes

Most compelling quote

Social proof, testimonial

Problem description in their words

Ad copy

Reaction to your solution pitch

Brand positioning

"I'd use it if..."

Key features to highlight

The interview template

Part 2

Interview structure, parts 1-3

Interview template

Part	Time	Goal
1. Opening	2 min	Frame it. Do not pitch.
2. Who are they?	2-3 min	Capture persona data for patterns
3. Problem exploration	5-7 min	The core: their reality

Interview structure, parts 4-6

Interview template

Part	Time	Goal
4. Current solutions	3-4 min	Past behavior. The money question.
5. Solution signal	3-4 min	Now pitch briefly. Watch them.
6. Wrap-up	2 min	Ask for referrals and follow-ups

Part 1: opening

Interview template

Say something like

"Hi, I'm [name]. I'm working on a class project at Columbia, exploring how people deal with [problem area]. I'm not selling anything. I just want to understand your experience. Would you mind chatting for about 25 minutes?"

Part 1: then invite honesty

Interview template

Then add

"Great. By the way, it's OK to be brutally honest with me. You can't hurt my feelings. That's how we improve our ideas."

Part 2: who are they?

Interview template

- "What do you do?" (student, job, field)
- "Do you run into [problem area]?"

Use it as an on-ramp: "How often do you cook?" leads to "What's the hardest part?"

Part 3: problem exploration

Interview template

01 Hardest part

"What's the hardest part about [problem context]?"

02 Last time

"Can you tell me about the last time that happened?"

03 Why

"Why was that hard?"

Part 3: go deeper

- Keep asking why. It can take 5 whys to reach the root cause.
- Follow up: "How often does this come up?" "How did that make you feel?"
- **Listen for emotion:** cursing, sighing, eye rolls, detailed rants.

If they shrug, the problem might not be real enough.

Part 4: current solutions

Interview template

Past behavior is the truth. This is your strongest signal.

01 **What you tried** "What, if anything, have you done to solve that problem?"

02 **What you don't love** "What don't you love about the solutions you've tried?"

Part 4: the money question

Interview template

Ask this

"Have you ever spent money trying to solve this? How much?"

- ✓ Even \$5 on an app counts. If they spent money, the problem is real.

Part 5: solution signal

Interview template

Now describe your solution, vaguely. Let them fill in the gaps.

Say something like

"So based on what you've told me, we're actually exploring building [one-sentence description]. The idea is [how it works, briefly]."

Part 5: then ask

- "Have you tried anything like this before?" If yes, why did they stop?
- "What's your initial reaction to that?"
- "Would this solve the problem we're discussing?"
- "What concerns do you have with this?"

Politely positive is not signal. "Wait, can I sign up?" is.

Part 6: wrap-up

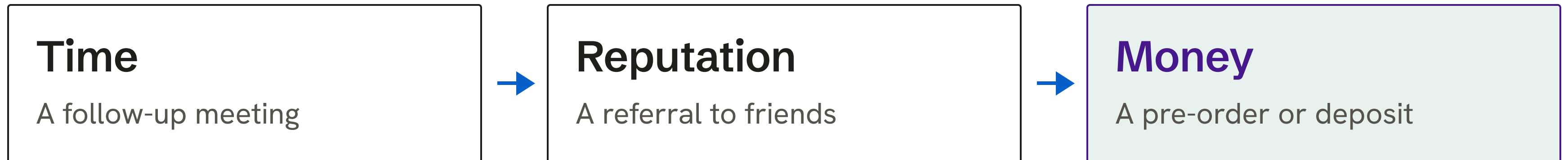
Interview template

- "If you could wave a magic wand and change one thing about this, what would it be?"
- "Is there anything about [problem] I should have asked but didn't?"
- "Do you know 1-2 other people who deal with this? Would you connect me?"
- "Would you be open to a 10-minute follow-up in a couple weeks?"
- "If you could pre-purchase this right now for \$10, would you?"

Currencies of commitment

Interview template

Each one is stronger than the last.



Signal cheat sheet

Interview template

✓ Strong signal

- "I HATE dealing with this," and has spent money
- Offers to introduce you to friends
- "Wait, can I sign up right now?"
- Names a specific price, or a specific task it would save time on

⊗ Weak signal

- Only agrees when you describe the problem
- "Yeah, that's kind of annoying," and a shrug
- "Let me know when it launches!" or "That's interesting"
- Has never tried to solve it or Googled it

How do I know when I'm done?

Interview template

The 3-of-5 rule

- Interview in batches of 5
- If 3 of 5 describe the same problem the same way, move forward
- No pattern by interview 5? Narrow your audience or look for adjacent problems.

The \$100 bet test

- Before your next interview, predict exactly what they'll say
- Would you bet \$100 you're right?
- If yes, you understand the problem. If not, keep interviewing.

**No signal? There
is probably an
adjacent problem.**

You're probably in the right neighborhood.

Finding adjacent problems

Interview template

Getting shrugs? Don't push harder. Explore the neighborhood.

01 Zoom out

"Forget [specific problem]. What's the most frustrating thing about [broader area]?"

02 Follow the energy

When something lights them up, go there. Ask all the same deep questions.

03 Walk their workflow

"Walk me through your typical [day, week, process]. Where do things break down?"

The adjacent problem: an example

Interview template

Your idea: a meal planning app for busy students.

01 Dead end

You ask about meal planning. "Eh, not really a problem. I just order Seamless." A shrug.

02 Zoom out

"What's the hardest part about feeding yourself?" "The money. I spend \$200 a week on delivery."

03 New idea

The pain isn't planning. It's what their workaround costs them.

A shrug isn't the end. Ask about the bigger area, and let them show you where the pain is.

What you'll do with your interviews

Part 2

Next week you'll synthesize what you heard. Here's what to listen for now.

Interview snapshot: what you learned

Homework

Fill it out within 1 hour of each interview. Memory fades fast.

Field	What to write
Who	Name or description
Problem confirmed?	Did they describe it unprompted?
Key quote	Most revealing thing, exact words
Emotional intensity	1 to 10

Interview snapshot: solutions and reaction

Homework

Field	What to write
Current solution	What are they doing now?
Money spent?	Have they paid for anything related?
Solution reaction	What did they say or do?
Biggest concern	What made them hesitate?

Interview snapshot: commitments and signal

Homework

Field	What to write
Referral?	Offered to connect you with others?
Follow-up?	Agreed to talk again?
Surprise	What contradicted your assumptions?
Signal	Green, yellow, or red

The 3-of-5 rule

**Did 3 of 5 describe
the same problem
the same way?**

You'll do 5 interviews per idea. Yes means you have signal. No means narrow your customer segment.

How to write your analysis

After the interviews

One short write-up per idea, after all 5 interviews for it.

01 Line them up

Put your 5 snapshots side by side.

02 Pull their words

Copy the pains people named, in their own words. Group the ones that match.

03 Count

How many of 5 described the same pain? What do they use today, and what does it cost them?

04 Decide

Keep, narrow, or drop this idea. Back it with your strongest quote.

Demo interview

Part 2

Watch how it works.

Watch for

Demo interview

- When do I ask about their life vs. mention my idea?
- When do I go deeper vs. move on?
- How do I handle silence?
- If signal is weak, how do I find the adjacent problem?

Your turn: practice

Part 2

Interview practice

Practice, 14 minutes

14:00

- 1 Pair up with someone from a different team.
- 2 Interview them about a real problem they have.
- 3 7 minutes each, then switch.
- 4 Use the five key questions on the next slide.
- 5 Listen for emotion: rants, specifics, money spent.

Interview practice: key questions

Practice

- 1 "What's the hardest part about [problem context]?"
- 2 "Can you tell me about the last time that happened?"
- 3 "Why was that hard?"
- 4 "What, if anything, have you done to solve that problem?"
- 5 "What don't you love about the solutions you've tried?"

Listening guide

Practice

While you practice, listen for:

- Is the problem told in the user's language, not yours?
- Is there emotional intensity, or just rational agreement?
- Did they mention spending money on the problem?
- Is a clear persona emerging, or is the audience too broad?

Homework

Homework

This week: 10 interviews per team

Homework

Per team

10

Interviews total, due
Friday, October 2

Ideas

2

Your team's top two,
from today

Per space

5

Interviews for each
problem space

What to submit by Friday, October 2

Homework

Due at the start of class. Add it to your team folder on GitHub, then paste the pull request link into Courseworks. We'll talk through what you heard.

- Your interview script
- Raw notes from every interview
- One interview snapshot per interview, filled out within 1 hour

Questions?

Go talk to ten people

Next class · Friday, October 2, 2026 · 1:10 to 3:40 PM · Pupin 420